



DEEP TECH INVESTOR DECK

The world model of industrial operations.

Minder learns what happens when people and machines act, across every plant it runs in.
Minder automates the industrial world.

LONDON, UNITED KINGDOM



THE THESIS

Industry's most valuable dataset has **never been recorded.**

What people decided, what the machines did next, and what it cost. Minder records it inside live plants, and trains a world model on it.

THE VISION

The autonomous factory runs itself.

LEVEL 1

See

One operational reality across people, machines and systems.

LEVEL 2

Remember

Every decision, fix and outcome kept as company memory.

LEVEL 3

Predict

The world model shows what each choice will change.

LEVEL 4

Act

The agent executes through approved tools. People approve what matters.

LEVEL 5

Run

Lines run themselves. Robots take the manual steps.

LIVE AT KASA

LIVE AT KASA

V0 · NOVEMBER 2026

2027

HORIZON

APPLICATIONS RUN LEVELS 1 AND 2 TODAY AND CAPTURE THE DATA · THE WORLD MODEL TAKES FACTORIES TO LEVELS 3 TO 5

THE BLIND SPOT

Systems record the plan. Plants run on unwritten rules.

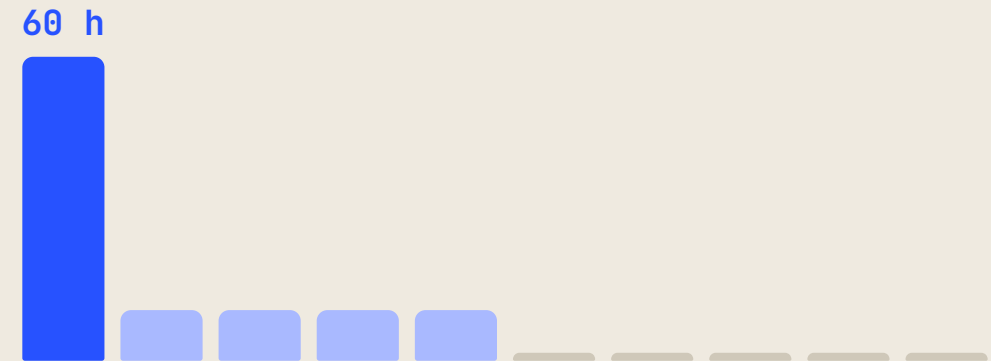
MRP ALLOCATION

100 hours across 10 operators.



OBSERVED ON THE FLOOR

One specialist carries 60 hours. Five wait.



Model the system, model the plan. The signal is in what no system records.

THE LANDSCAPE

Every model family learns a different unit of the world.

LANGUAGE	Text	<i>What does this mean?</i>	Frontier LLMs
TIME SERIES	Signals, no actions	<i>What comes next?</i>	TimesFM, Chronos
VIDEO WORLD	Pixels	<i>What will I see?</i>	Genie, Cosmos
ROBOTICS	Motion, one body	<i>How do I move?</i>	π 0, Skild
INDUSTRIAL SUITES	The historian	<i>What is the asset doing?</i>	Siemens, Cognite
MINDER	Decisions and consequences	<i>What happens if we do this?</i>	Plant scale

THE LEARNING UNIT

A decision, followed to its consequence.

$p(s', y | c, s, a)$

Learn it across plants. Adapt it to a new site with days of data, not years.

c

CONTEXT

Plant, line,
operating model,
product, weather

s

STATE

Signals, alarms,
queues, work in
progress

a

ACTION

Operator decision
or approved
command

s'

NEXT STATE

The measured
response

y

OUTCOME

Quality, time,
energy, safety

e

WHY

Reason,
confidence,
diagnosis trace

STRUCTURE FIRST

A machine is logic before it is data.

Electrical, mechanical, pneumatic and hydraulic subsystems pass signals through one cycle. Product and environment move everything else.

MACHINE LOGIC

Cycle graphs from PLC signals, manuals and drawings

A fault that looks mechanical is often a late encoder signal.



PRODUCT AND ENVIRONMENT

Load, mix, moisture, ambient heat and humidity as conditioning

A humid afternoon can spoil a batch two weeks later.



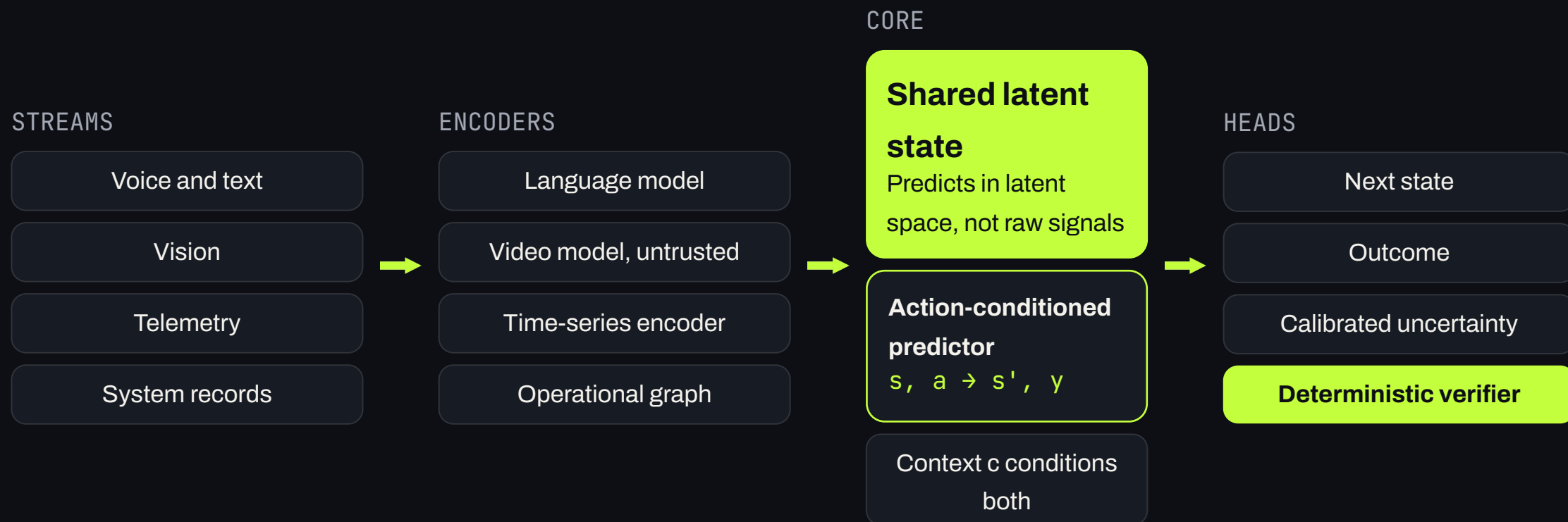
LEARNED DYNAMICS

Residuals the logic cannot explain, learned across the fleet

Less data per site. Physically consistent predictions.

Hybrid by design: structure from machine logic, dynamics from data.

One latent state of the plant, predicted under action.



v0 fine-tunes NVIDIA Cosmos on Google Cloud, trained on the first-party event log. Served to the agent; every action gated by machine logic and human approval.

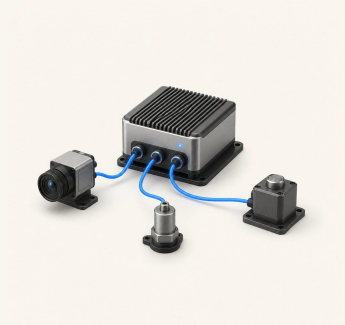
THE DATA ENGINE

Every deployment is a sensor for the model.



PEOPLE

Voice in their own language: the decision and the reason. Live at Kasa in Vietnamese.



MACHINES AND VISION

Retrofit sensors and cameras at \$120–300 a device. PLC signals read only.



SYSTEMS

ERP, MES, CMMS and historians, read only by default.



DEMONSTRATIONS • RESEARCH

Human demonstrations become trajectories for embodied models.

DEPLOY FIRST WHERE

Real alternatives

The answer is tacit

Outcomes are
observable

Decisions recur

THE LONG TAIL

Capture the rare event before it happens.

A fault appears once every 18 months. The one person who knows the fix is on holiday.

01 • POOLING

Rare in one plant, routine across a fleet.

02 • ELICITATION

In quiet hours Minder asks experts what they would do if...
Answers become labelled counterfactuals.

03 • DIAGNOSIS TRACES

Structured fault finding, logged step by step, becomes the explanation label.

The model learns failure modes **no single plant has seen yet.**

Does one model transfer across plants?

NULL HYPOTHESIS	Every factory needs its own bespoke model.
ASSET CLASS	Production assets at Hòa Phát plants first; Kasa’s industrial dryers as a second domain.
SIGNALS	Operator actions, process telemetry, ambient conditions, energy and quality.
TEST	Fine-tune NVIDIA Cosmos on plants A and B. Adapt to plant C with two weeks of data. First run at a Hòa Phát plant in November 2026.
PASS	Transfer beats a model trained on each plant’s own history, with calibrated uncertainty, on five more plants by March 2027.
OUTPUT	Paper and open benchmark: FactoryTraj-Bench.

Each stage is gated by evidence.

V0

Cross-site

One asset class, several plants.

GATE · BEATS
SITE-TRAINED
BASELINE

V1

Cross-asset

Compressors and cold storage in food plants.

GATE · TRANSFERS
ACROSS ASSET CLASSES

V2

Plant-level

Orders, people and machines together. What is our real capacity?

GATE ·
COUNTERFACTUALS HOLD
ON LOGGED
INTERVENTIONS

V3

Supervised action

Proof-carrying actions through validated adapters.

GATE · ZERO UNSAFE
ACTIONS IN SHADOW
MODE

HORIZON

Embodiment

Robots trained on Minder trajectories.

GATE · CAPABILITY
PROVEN ON THE LINE

SCORED ON: TRANSFER · COUNTERFACTUAL VALIDITY · CALIBRATION · ABSTENTION

WHY NOW

Four curves have crossed.

<62 h

of action-conditioned robot video, on top of passive video, let V-JEPA 2 plan zero-shot.

51M

datapoints across 23K executions in FactoryNet, with early cross-embodiment transfer.

Any language

Voice models capture shop-floor decisions in the worker's own language.

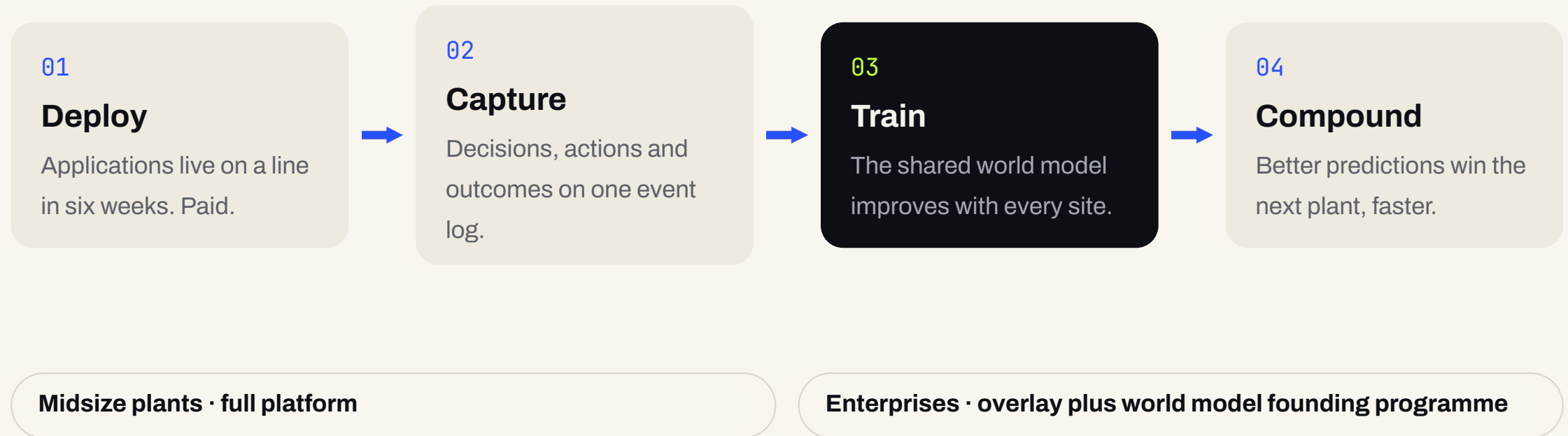
Cosmos 3

Open world foundation model that predicts actions and future states, now on edge devices. Our v0 starting point.

THE FLYWHEEL

Applications pay for the fleet. The fleet trains the model.

Tesla sold cars that paid for themselves, then trained self-driving on the fleet. Minder does the same for industry.





TRACTION • THE ENGINE IS RUNNING

Live and paid in a plant processing 20 tonnes a day.

7

departments on one
event log

30/30

assigned workers active

24

month signed contract

6 wks

to go live, two FDEs

39,117

events logged per day

1 + 8

paid site plus signed MOUs

WORLD MODEL PARTNER • IN DISCUSSION

Hòa Phát Group • Nestlé, Isuzu

WORLD MODEL PARTNER

Hòa Phát is our first world model partner.

01 ~30 factories

Vietnam's largest steelmaker: operations data across a national group of steel and appliance plants.

02 Full suite

Applications and agent deployed plant by plant, on founding terms.

03 v0 · November 2026

NVIDIA Cosmos fine-tuned on their operations data, trained on Google Cloud.

04 Their AI team, our FDEs

Trained by us, paid by Hòa Phát. We keep the weights, code and training rights.

WHAT COMPOUNDS

Four advantages that grow with every site.

01	Data no one else captures	Decisions and consequences, recorded at the source, in the worker's language.
02	Rights by contract	Customers own raw data. Anonymised, derived data trains the shared model.
03	Cheaper every site	First site: six to eight FDE weeks. Target for repeat sites: two to three.
04	Verifiable autonomy	Every action leaves an independently verifiable receipt.

BUSINESS MODEL

Applications now. Licences next. A model API after.

NOW

Applications

Site licence plus value share. Live at Kasa on a 24-month contract.

NEXT

Enterprise overlay

Per plant, from April 2027, on the customer's existing stack. Founding partners first.

LATER

World model API

Machine builders, software vendors, robot makers and insurers.

Hardware \$120–300 per device

Cloud and inference ~\$12k per plant a year

Gross margin 70%+ entry, 90%+ enterprise

MARKET • TAM, SAM, SOM

\$100bn a year to win in three regions. China is upside.

TAM

\$100bn a year

10% more capacity on ~\$5tn of output,
20% captured by Minder.

SAM

\$39bn a year

246,600 plants: applications \$26bn plus
world model \$13bn.

SOM

\$390m a year

About 2,500 plants within 5 to 7 years,
starting with Hòa Phát's 30 factories.

Southeast Asia \$67k to \$225k per plant

Europe, East Asia \$135k to \$449k per plant

Per plant applications plus world model at 50%

TEAM

Built by people who can make it happen.

CO-FOUNDER / CEO

Eric Nguyen

Founders for Founders s16vc + AltaIR Capital AltaLab

Field deployment, commercial motion and partnerships.
Led the Kasa deployment in 6 weeks, founded Strand Ventures (UK's first student VC).

B2B GTM / finance / investor relations / operation / growth

CO-FOUNDER / CTO

Bill Nguyen

Silicon Valley Fellow + Founders for Founders s16vc

Built an open-source AI agent for Snapchat Spectacles smart glasses. Built Jan v3.5 (400k downloads) & Byterover CLI (5k starred)

AI research / product / users / open source / engineering

OUR TEAM OF 21, FOUNDERS INCLUDED



CAPITAL PLAN

\$5M to train and scale the model.

\$5M raising now

MILESTONES THE CAPITAL BUYS

- ✓ Nov 2026 · first model tested at a Hòa Phát plant
- ✓ Mar 2027 · live on 5 more plants, beating their baselines
- ✓ Dec 2027 · world model across at least 20 plants
- ✓ Sep 2028 · world model running at 50 to 100 plants

USE OF FUNDS

World model team and evaluation	29.1%
Platform, agent and inference	22.1%
G&A, security and legal	15.1%
GPU compute and data	15.0%
Forward deployment	11.0%
Go to market	7.7%

World model: 44% of planned spend.

SEIS/EIS advance assurance in preparation · UK topco
Celesnity Ltd

BACK THE WORLD MODEL OF INDUSTRY

Every plant that runs Minder teaches the world model.



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